Files

- Files are named locations on disk to store related information.
- They are used to permanently store data in non-volatile memory (e.g., hard disk).

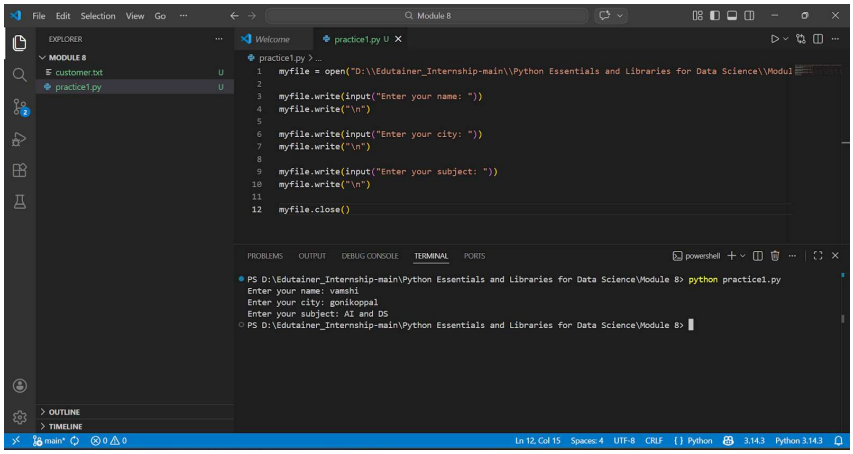
Using Python's File Input Output capability we can:

- Open a file
- Read or write a file
- Close a file

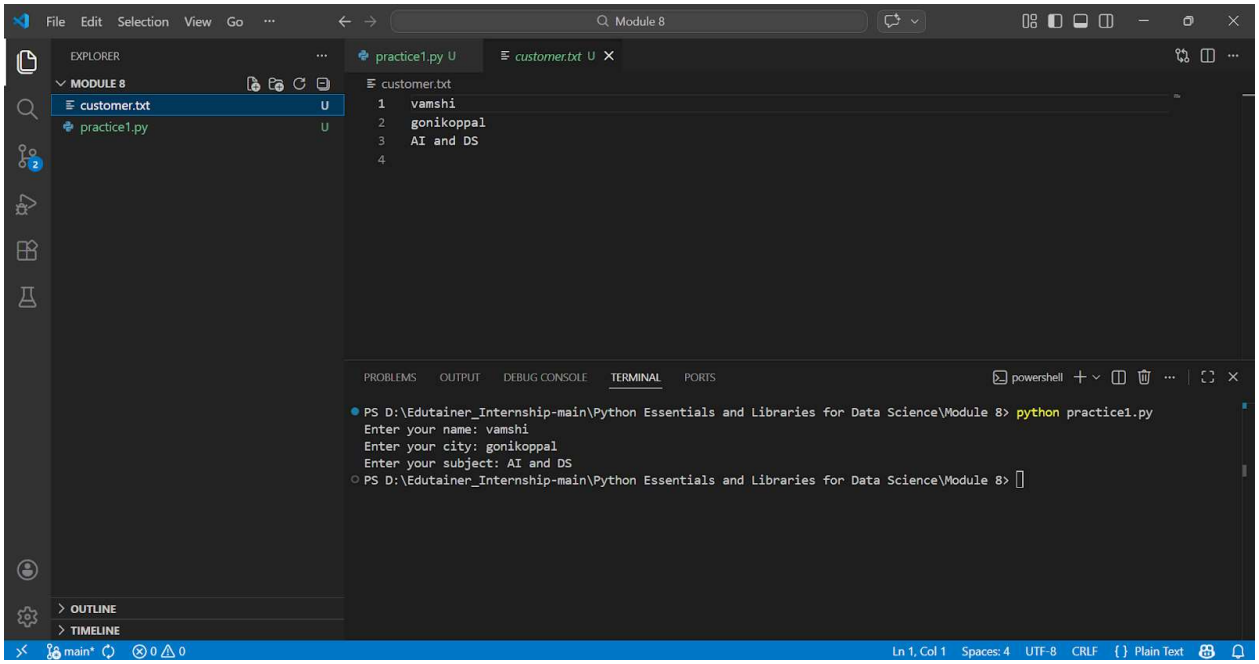
A file can be opened in one of the following modes:

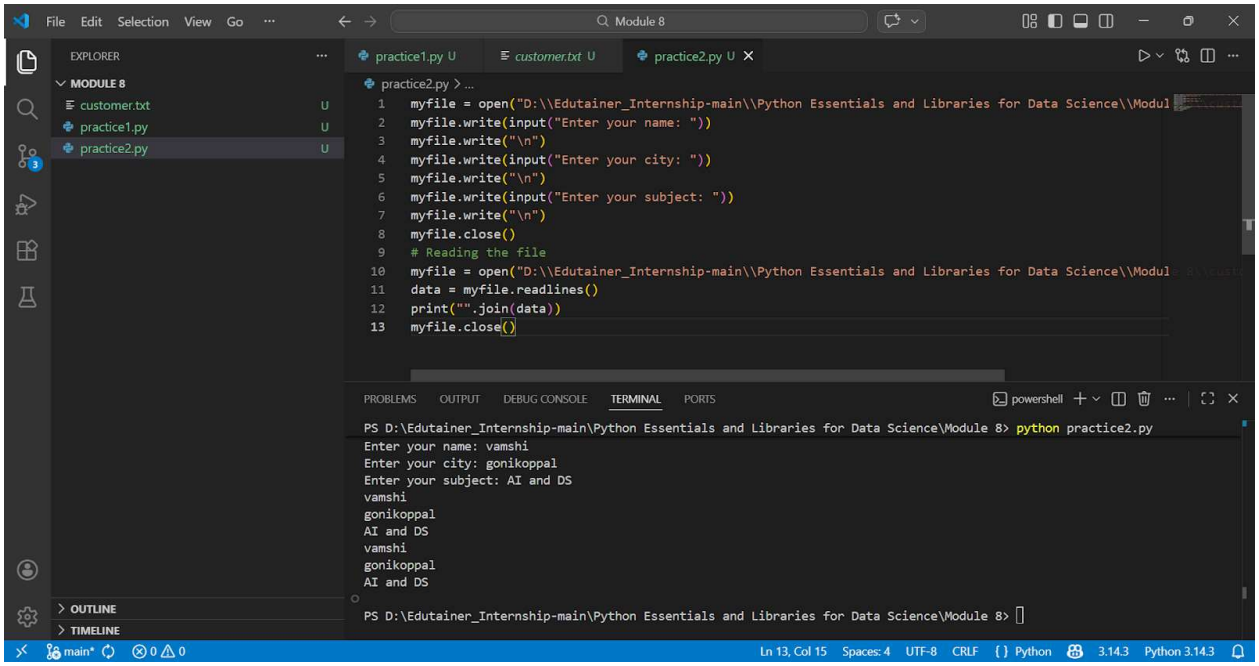
- read
- write
- read and write
- append

Note:
File handling has limited capabilities and should not be compared with operations that are done with database system support as backend.



A customer.txt file is created….

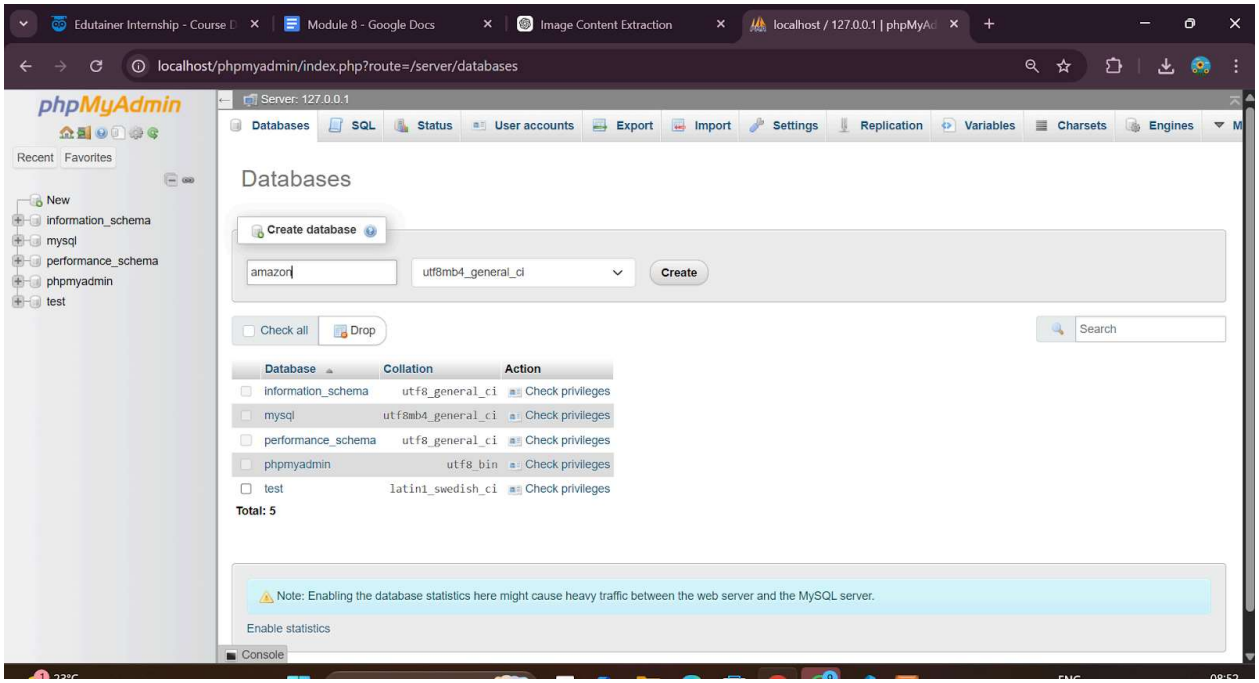


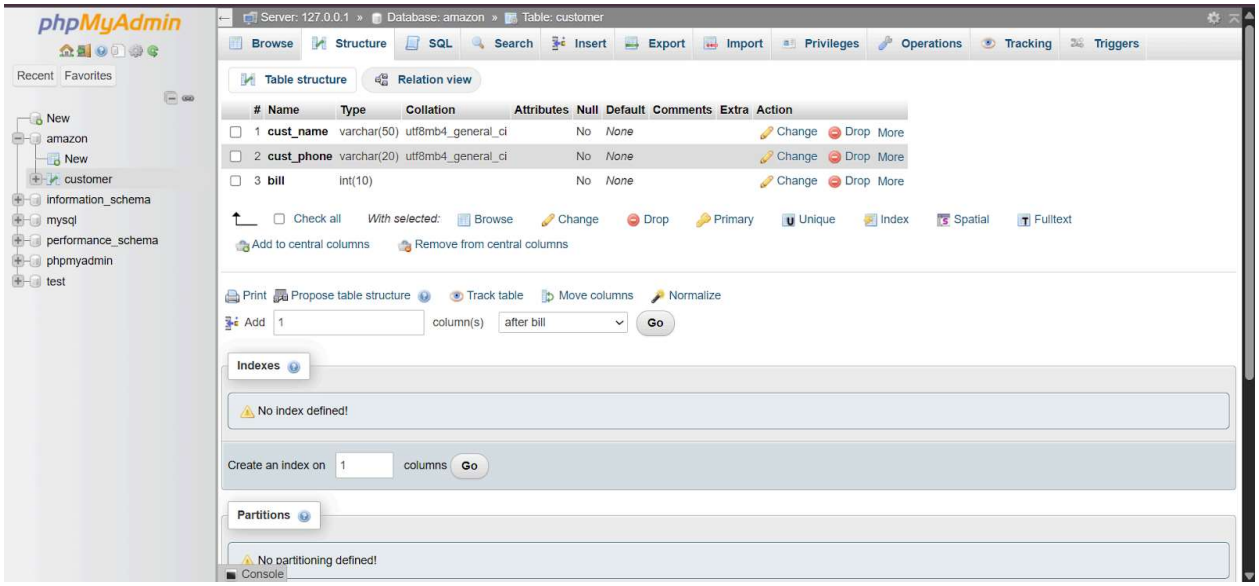


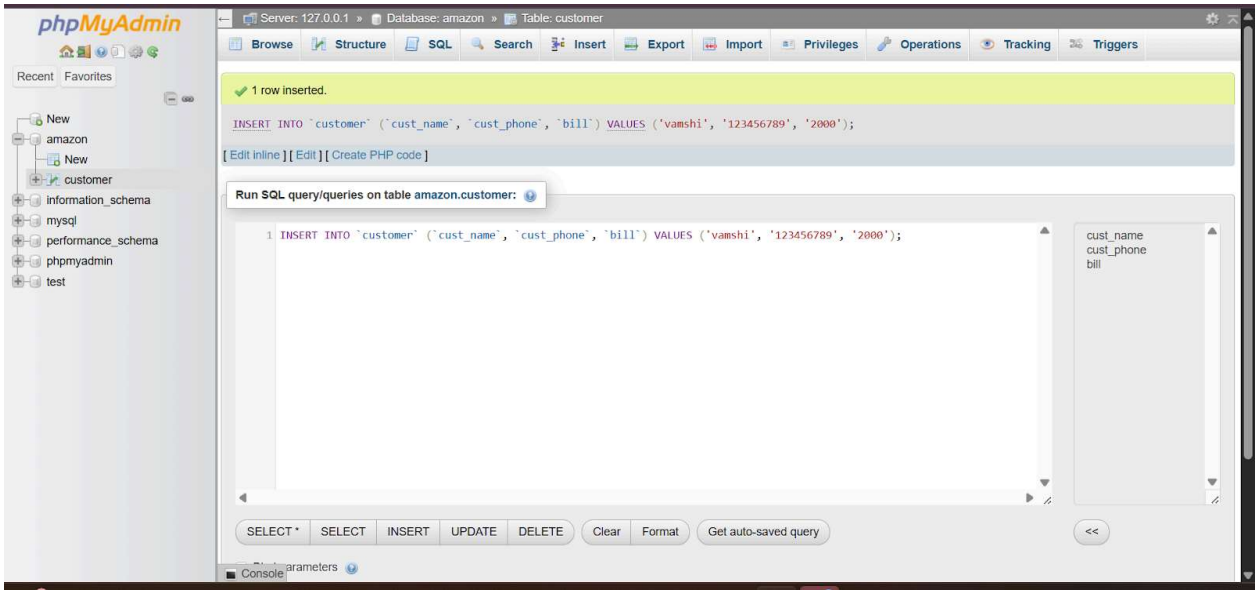
Steps for Database Connectivity

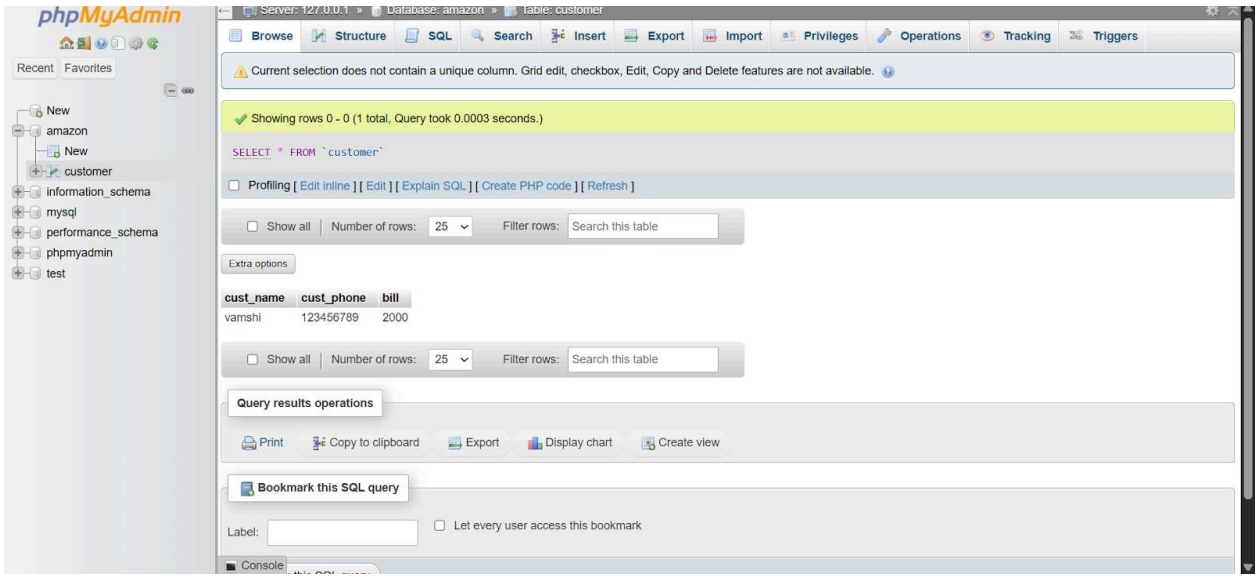
- Download and install XAMPP
- Create database
- Create table
- Insert data into the table and verify

- Module for database connectivity
- pip install pymysql
- Import the module in the program







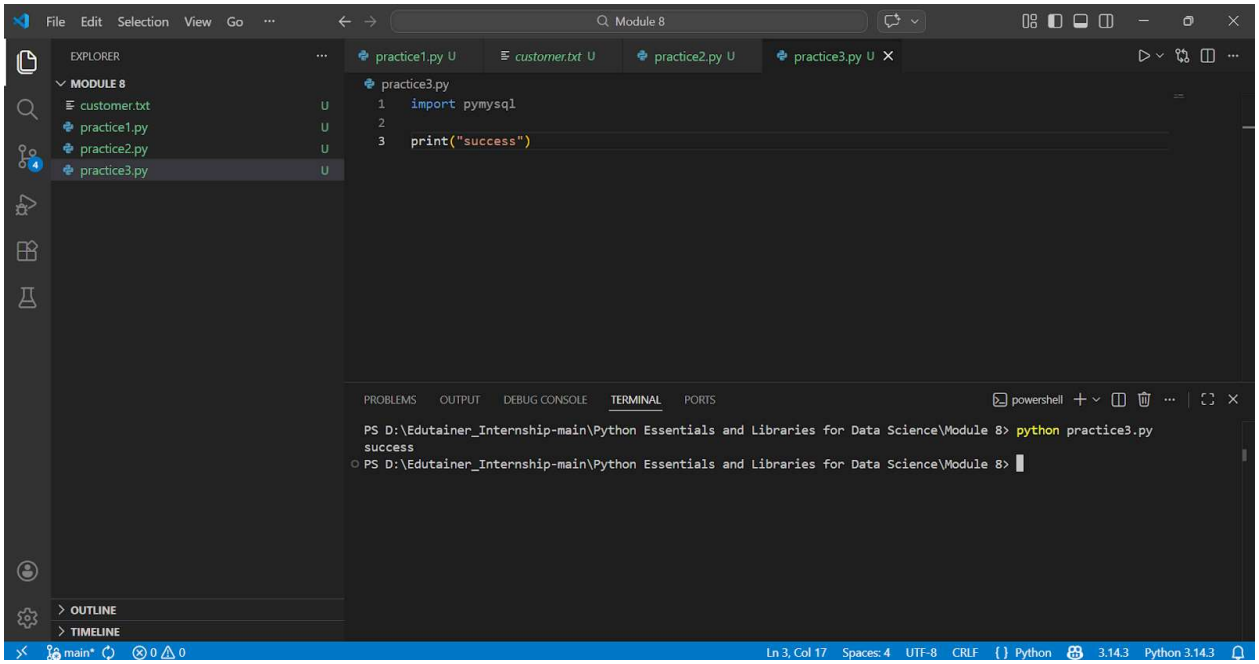


Go to command prompt

Type:-pip install pymysql

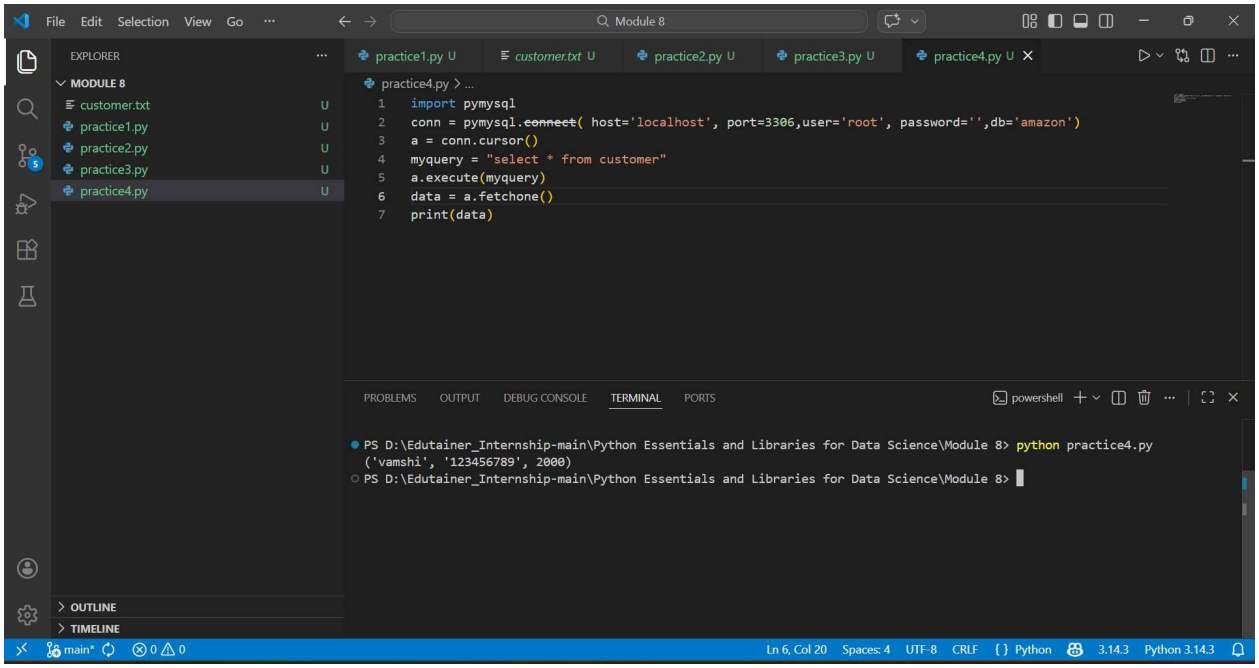
To install mysql

Check whether its installed through this program



Fetching data from database

1.Fetching information of one user.



2.Fetching information of all users.

```python
import pymysql
conn = pymysql.connect( host='localhost', port=3306,user='root', password='',db='amazon')
a = conn.cursor()
myquery = "select * from customer"
a.execute(myquery)
data = a.fetchall()
print(data)
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8> python practice5.py
(('vamshi', '123456789', 2000), ('jithin', '9876543221', 1000))
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8>
```

Fetching the customers using where clause

```python
import pymysql
conn = pymysql.connect(host='localhost',port=3306,user='root',password='',db='amazon')
a = conn.cursor()
client = input("Enter customer name to search: ")
sql = "select * from customer where cust_name = '" + client + "'"
print(sql)
a.execute(sql)
data = a.fetchall()

for item in data:
    print(item)
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8> python practice6.py
Enter customer name to search: vamshi
select * from customer where cust_name = 'vamshi'
('vamshi', '123456789', 2000)
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8> python practice6.py
Enter customer name to search: siya
select * from customer where cust_name = 'siya'
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8>
```

Inserting new values:-

```python
import pymysql
conn = pymysql.connect(host='localhost',port=3306,user='root',password='',db='amazon')
a = conn.cursor()
client = input("Enter customer name: ")
phone = input("Enter customer phone: ")
bill_amt = input("Enter customer bill amt: ")
sql = "insert into customer values('" + client + "','" + phone + "','" + bill_amt + "')"
print(sql)
a.execute(sql)
conn.commit()
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8> python practice7.py
Enter customer name: sachi
Enter customer phone: 5647382910
Enter customer bill amt: 4000
insert into customer values('sachi','5647382910','4000')
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8>
```

Server: 127.0.0.1 » Database: amazon » Table: customer

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)

```
SELECT * FROM `customer`
```

cust_name	cust_phone	bill
vamshi	123456789	2000
jithin	9876543221	1000
sachi	5647382910	4000

Deleting a particular customer record:-

```python
import pymysql
conn = pymysql.connect(host='localhost',port=3306,user='root',password='',db='amazon')
a = conn.cursor()
client = input("Enter customer name to delete: ")
sql = "delete from customer where cust_name = '" + client + "'"
print(sql)
a.execute(sql)
conn.commit()
```

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8> python practice8.py
Enter customer name to delete: jithin
delete from customer where cust_name = 'jithin'
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8>
```

Server: 127.0.0.1 » Database: amazon » Table: customer

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0002 seconds.)

SELECT * FROM `customer`

cust_name	cust_phone	bill
vamshi	123456789	2000
sachi	5647382910	4000

Updating value of specific record:-

practice9.py

```python
import pymysql
conn = pymysql.connect(host='localhost',port=3306,user='root',password='',db='amazon')
a = conn.cursor()
client = input("Enter customer name you want to update: ")
sql = "update customer set bill = 70000 where cust_name = '" + client + "'"
print(sql)
a.execute(sql)
conn.commit()
```

Terminal output:

```
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8> python practice9.py
Enter customer name you want to update: vamshi
update customer set bill = 70000 where cust_name = 'vamshi'
PS D:\Edutainer_Internship-main\Python Essentials and Libraries for Data Science\Module 8>
```

Server: 127.0.0.1 » Database: amazon » Table: customer

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 1 (2 total, Query took 0.0003 seconds.)

SELECT * FROM `customer`

cust_name	cust_phone	bill
vamshi	123456789	70000
sachi	5647382910	4000